Database Systems - SE3060
L06_xml-Examples and Lab Sheet 07 (Answers)
Year 3 – SE Batch (WD)

```
-- Example: Untyped XML Column in Table
CREATE TABLE AdminDocs (
id int  primary key,
xDoc Xml not null
) -- Example: Inserting Data into Untyped XML Column
INSERT INTO AdminDocs VALUES (6,
'<catalog>
<product dept="WMN">
<number>557</number>
<name language="en">Fleece Pullover</name>
<colorChoices>navy black</colorChoices>
</product>
<product dept="ACC">
<number>563</number>
<name language="en">Floppy Sun Hat</name>
</product>
<product dept="ACC">
<number>443</number>
<name language="en">Deluxe Travel Bag</name>
</product>
<product dept="MEN">
<number>784</number>
<name language="en">Cotton Dress Shirt</name>
<colorChoices>white gray</colorChoices>
<desc>Our <i>favorite</i> shirt!</desc>
</product>
</catalog>')
INSERT INTO AdminDocs VALUES (2,
'<doc id="123">
<sections>
```

```
<section num="1"><title>XML Schema</title></section>
<section num="3"><title>Benefits</title></section>
<section num="4"><title>Features</title></section>
</sections>
</doc>')
select *
from AdminDocs -- Example: Using Query() Method
SELECT id, xDoc.query('/catalog')
FROM   AdminDocs
SELECT id, xDoc.query('/catalog/product')
FROM   AdminDocs
SELECT id, xDoc.query('//product')
FROM   AdminDocs
SELECT id, xDoc.query('/*/product')
FROM   AdminDocs
SELECT id, xDoc.query('/*/product[@dept="WMN"]')
FROM   AdminDocs

SELECT id, xDoc.query('/*/child::product[attribute::dept="WMN"]')
FROM   AdminDocs

SELECT id, xDoc.query('/*/child::product[attribute::dept="WMN"]')
FROM   AdminDocs

SELECT id, xDoc.query('//product[dept="WMN"]')
FROM   AdminDocs

SELECT id, xDoc.query('descendant-or-self::product[attribute::dept="WMN"]')
FROM   AdminDocs
```

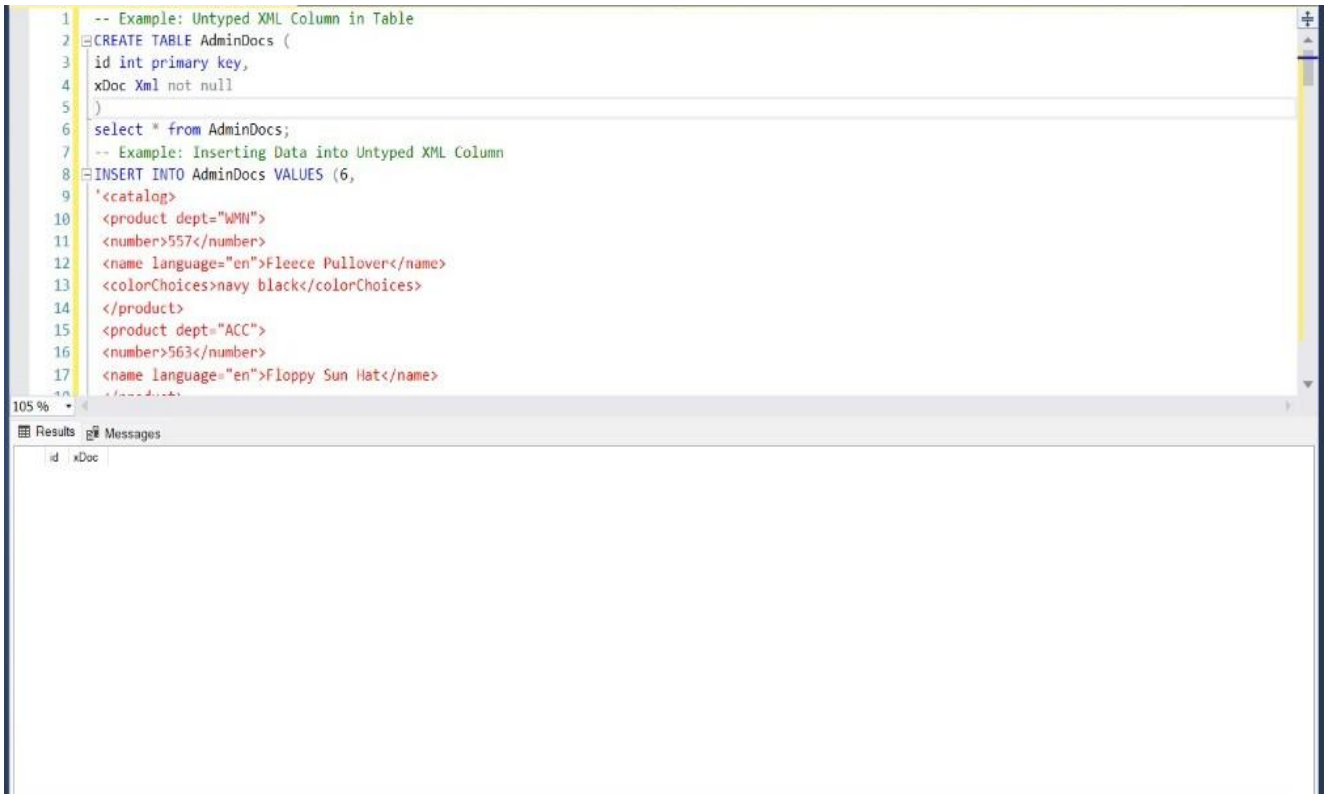
```
SELECT id, xDoc.query('//product[number > 500]')
FROM   AdminDocs
where id=6

SELECT id, xDoc.query('//product/number[. gt 500]')
FROM   AdminDocs
where id=6

SELECT id, xDoc.query('/catalog/product[4]')
FROM   AdminDocs
where id=6

SELECT id, xDoc.query('//product[number > 500][@dept="ACC"]')
FROM   AdminDocs
where id=6

SELECT id, xDoc.query('//product[number > 500][1]')
FROM   AdminDocs
where id=6

SELECT xDoc.query(' for $prod in //product
        let $x:=$prod/number
        return $x')
FROM   AdminDocs
where id=6

SELECT xDoc.query(' for $prod in //product
       let $x:=$prod/number
       where $x>500
```

```
    return $x')
FROM   AdminDocs
where id=6

SELECT xDoc.query(' for $prod in //product
   let $x:=$prod/number
   return $x')
FROM   AdminDocs
where id=6

SELECT xDoc.query('for $prod in //product
   let $x:=$prod/number
   where $x>500
   return (<Item>{$x}</Item>)')
FROM   AdminDocs
where  id=6

SELECT xDoc.query(' for $prod in //product[number > 500]
   let $x:=$prod/number
   where $x>500
   return (<Item>{$x}</Item>)')
FROM   AdminDocs
where  id=6

SELECT xDoc.query('for $prod in //product
   let $x:=$prod/number
   where $x>500
   return (<Item>{data($x)}</Item>)')
FROM   AdminDocs
where  id=6
```

```
SELECT xDoc.query('for $prod in //product
    let $x:=$prod/number
    return  if  ($x>500)
    then <book>{$x}</book>
    else <paper>{$x}</paper>')
FROM   AdminDocs
where  id=6

SELECT xDoc.query('for $prod in //product
    let $x:=$prod/number
    return  if  ($x>500)
    then <book>{data($x)}</book>
    else <paper>{data($x)}</paper>')
FROM   AdminDocs
where  id=6

 -- Example: Using Exist() Method

SELECT id
FROM   AdminDocs
WHERE  xDoc.exist ('/doc[@id = 123]') = 1

 --Example: Using Value() Method

SELECT xDoc.value('(/doc//section[@num = 3]/title)[1]', 'varchar(100)')
FROM   AdminDocs
 --Example: Insertion of Subtree into XML Instances
select * from AdminDocs where id=2
```

```
UPDATE AdminDocs SET xDoc.modify('
  insert
   <section num="2">
       <title>Background</title>
   </section>
  after (/doc//section[@num=1])[1]')

UPDATE AdminDocs SET xDoc.modify('
  delete
//section[@num="2"]')
```

Screenshots



```
22  <number>443</number>
23  <name language="en">Deluxe Travel Bag</name>
24  </product>
25  <product dept="MEN">
26  <number>784</number>
27  <name language="en">Cotton Dress Shirt</name>
28  <colorChoices>white gray</colorChoices>
29  <desc>Our <i>favorite</i> shirt!</desc>
30  </product>
31 </catalog>')
32 INSERT INTO AdminDocs VALUES (2,
33  '<doc id="123">
34  <sections>
35  <section num="1"><title>XML Schema</title></section>
36  <section num="3"><title>Benefits</title></section>
```

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Results | Messages

	id	xDoc
1	2	<doc id="123"><sections><section num="1"><title>XML Schema</title></section><section nu...
2	6	<catalog><product dept="WMN"><number>557</number><name language="en">Fleece Pull...

```
31 </catalog>')
32 INSERT INTO AdminDocs VALUES (2,
33  '<doc id="123">
34   <sections>
35   <section num="1"><title>XML Schema</title></section>
36   <section num="3"><title>Benefits</title></section>
37   <section num="4"><title>Features</title></section>
38   </sections>
39  </doc>')
40 select *
41  from AdminDocs
42  -- Example: Using Query() Method
43 SELECT id, xDoc.query('/catalog')
44  FROM AdminDocs
45 SELECT id, xDoc.query('/catalog/product')
46  FROM AdminDocs
47 SELECT id, xDoc.query('//product')
48  FROM AdminDocs
49 SELECT id, xDoc.query('/*/product')
50  FROM AdminDocs
```

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Results | Messages

	id	(No column name)
1	2	
2	6	<catalog><product dept="WMN"><number>557</number...

```
31  </catalog>')
32  INSERT INTO AdminDocs VALUES (2,
33  '<doc id="123">
34   <sections>
35   <section num="1"><title>XML Schema</title></section>
36   <section num="3"><title>Benefits</title></section>
37   <section num="4"><title>Features</title></section>
38   </sections>
39  </doc>')
40  select *
41  from AdminDocs
42  -- Example: Using Query() Method
43  SELECT id, xDoc.query('/catalog')
44  FROM AdminDocs
45  SELECT id, xDoc.query('/catalog/product')
46  FROM AdminDocs
47  SELECT id, xDoc.query('//product')
48  FROM AdminDocs
49  SELECT id, xDoc.query('/*/product')
50  FROM AdminDocs
```

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Results | Messages

	id	(No column name)
1	2	
2	6	<product dept="WMN"><number>557</number><name la...

```
37   <section num="4"><title>Features</title></section>
38   </sections>
39  </doc>')
40  select *
41  from AdminDocs
42  -- Example: Using Query() Method
43  SELECT id, xDoc.query('/catalog')
44  FROM AdminDocs
45  SELECT id, xDoc.query('/catalog/product')
46  FROM AdminDocs
47  SELECT id, xDoc.query('//product')
48  FROM AdminDocs
49  SELECT id, xDoc.query('/*/product')
50  FROM AdminDocs
51  SELECT id, xDoc.query('/*/product[@dept="WMN"]')
52  FROM AdminDocs
53  SELECT id, xDoc.query('/*/child::product[attribute::dept="WMN"]')
54  FROM AdminDocs
55  SELECT id, xDoc.query('/*/child::product[attribute::dept="WMN"]')
56  FROM AdminDocs
```

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Results | Messages

	id	(No column name)
1	2	
2	6	<product dept="WMN"><number>557</number><name la...

	id	(No column name)
1	2	
2	6	<product dept="WMN"><number>557</number><name la...

	id	(No column name)
1	2	
2	6	<product dept="WMN"><number>557</number><name la...

```
67 SELECT id, xDoc.query('/catalog/product[4]')
68 FROM AdminDocs
69 where id=6
70 SELECT id, xDoc.query('//product[number > 500][@dept="ACC"]')
71 FROM AdminDocs
72 where id=6
73 SELECT id, xDoc.query('//product[number > 500][1]')
74 FROM AdminDocs
75 where id=6
76 SELECT xDoc.query(' for $prod in //product
77 let $x:=$prod/number
78 return $x')
79 FROM AdminDocs
80 where id=6
81 SELECT xDoc.query(' for $prod in //product
82 let $x:=$prod/number
83 where $x>500
84 return $x')
```

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Results | Messages

	id	(No column name)
1	6	<number>557</number><number>563</number><number>...

	id	(No column name)
1	6	<product dept="MEN"><number>784</number><name la...

	id	(No column name)
1	6	<product dept="ACC"><number>563</number><name la...

	id	(No column name)
1	6	<product dept="WMN"><number>557</number><name la...

```
73 SELECT id, xDoc.query('//product[number > 500][1]')
74 FROM AdminDocs
75 where id=6
76 SELECT xDoc.query(' for $prod in //product
77 let $x:=$prod/number
78 return $x')
79 FROM AdminDocs
80 where id=6
81 SELECT xDoc.query(' for $prod in //product
82 let $x:=$prod/number
83 where $x>500
84 return $x')
85 FROM AdminDocs
86 where id=6
87 SELECT xDoc.query(' for $prod in //product
88 let $x:=$prod/number
89 return $x')
90 FROM AdminDocs
```

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Results | Messages

	(No column name)
1	<number>557</number><number>563</number><number>4...

	(No column name)
1	<number>557</number><number>563</number><number>7...

```
let $x:=$prod/number
return $x')
FROM AdminDocs
where id=6
SELECT xDoc.query('for $prod in //product
let $x:=$prod/number
where $x>500
return (<Item>{$x}</Item>)')
FROM AdminDocs
where id=6
SELECT xDoc.query(' for $prod in //product[number > 500]
let $x:=$prod/number
where $x>500
return (<Item>{$x}</Item>)')
FROM AdminDocs
where id=6
SELECT xDoc.query('for $prod in //product
let $x:=$prod/number
```

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Results | Messages

	(No column name)
1	<number>557</number><number>563</number><number>4...

	(No column name)
1	<Item><number>557</number></Item><Item><number>56...

	(No column name)
1	<Item><number>557</number></Item><Item><number>56...

```
FROM AdminDocs
where id=6
SELECT xDoc.query('for $prod in //product
let $x:=$prod/number
return if ($x>500)
then <book>{data($x)}</book>
else <paper>{data($x)}</paper>')
FROM AdminDocs
where id=6
-- Example: Using Exist() Method
SELECT id
FROM AdminDocs
WHERE xDoc.exist ('/doc[@id = 123]') = 1
--Example: Using Value() Method
SELECT xDoc.value('(/doc//section[@num = 3]/title)[1]', 'varchar(100)')
FROM AdminDocs
--Example: Insertion of Subtree into XML Instances
select * from AdminDocs where id=2
```

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Results | Messages

	(No column name)
1	<Item>557</Item><Item>563</Item><Item>784</Item>

	(No column name)
1	<book><number>557</number></book><book><number>56...

	(No column name)
1	<book>557</book><book>563</book><paper>443</paper...

```sql
-- Example: Using Exist() Method
SELECT id
FROM AdminDocs
WHERE xDoc.exist ('/doc[@id = 123]') = 1
--Example: Using Value() Method
SELECT xDoc.value('(/doc//section[@num = 3]/title)[1]', 'varchar(100)')
FROM AdminDocs
--Example: Insertion of Subtree into XML Instances
select * from AdminDocs where id=2
UPDATE AdminDocs SET xDoc.modify('
 insert
 <section num="2">
 <title>Background</title>
 </section>
 after (/doc//section[@num=1])[1]')
UPDATE AdminDocs SET xDoc.modify('
 delete
 //section[@num="2"]')
```

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Results | Messages

	id
1	2

	(No column name)
1	Benefits
2	NULL

	id	xDoc
1	2	<doc id="123"><sections><section num="1"><title>X...

```sql
WHERE xDoc.exist ('/doc[@id = 123]') = 1
--Example: Using Value() Method
SELECT xDoc.value('(/doc//section[@num = 3]/title)[1]', 'varchar(100)')
FROM AdminDocs
--Example: Insertion of Subtree into XML Instances
select * from AdminDocs where id=2
UPDATE AdminDocs SET xDoc.modify('
 insert
 <section num="2">
 <title>Background</title>
 </section>
 after (/doc//section[@num=1])[1]')
UPDATE AdminDocs SET xDoc.modify('
 delete
 //section[@num="2"]')
```

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Messages

```
(2 rows affected)

(2 rows affected)

Completion time: 2025-09-14T22:50:52.2420715+05:30
```

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